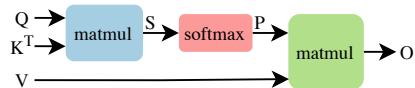


$$\text{softmax}(QK^T)V$$



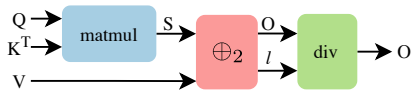
```

for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
  for all  $i \in [0, N)$  do
    for  $j \in [0, N)$  do
       $\tilde{m}_i := \max(m_i, S_{i,j})$ 
    for all  $i \in [0, N)$  do
      for  $j \in [0, N)$  do
         $P_{i,j} := \exp(S_{i,j} - \tilde{m}_i)$ 
         $\ell_i := \ell_i + P_{i,j}$ 
         $P_{i,j} := P_{i,j} / \ell_i$ 
  for all  $i, j \in [0, N) \times [0, d)$  do
    for  $k \in [0, N)$  do
       $O_{i,j} := O_{i,j} + P_{i,k}V_{k,j}$ 

```

dependent reductions make
j-loop unfusable

$$\text{fused_softmax_matmul}(QK^T, V)$$



```

for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 

```

```

for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do

```

$$\tilde{P} = e^{S_{i,j}}$$

$$\begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} := \begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} \oplus_2 \begin{bmatrix} S_{i,j} \\ \tilde{P} \\ \tilde{P}V_j \end{bmatrix}$$

```

for all  $i, j \in [0, N) \times [0, N)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 

```

fuse $S_{i,j}$ producer into $\oplus_2$
 fuse $O_{i,j}$ consumer into $\oplus_2$
 tile by (B_r, B_c)
 decomposeOp

```

for all  $i \in [0, \lfloor \frac{N}{B_r} \rfloor)$  do
  for  $j \in [0, \lfloor \frac{N}{B_c} \rfloor)$  do
    for all  $i_t \in [0, B_r)$  do
      for  $j_t \in [0, B_c)$  do

```



```

for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 

```

```

for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do

```

$$\tilde{m} = \max(m_i, S_{i,j})$$

$$\tilde{P} = e^{S_{i,j} - \tilde{m}}$$

$$\tilde{\alpha} = e^{m_i - \tilde{m}}$$

$$m_i := \tilde{m}$$

$$\ell_i := \alpha \ell_i + \tilde{P}$$

```

for all  $k \in [0, d)$  do
   $O_{i,k} := \alpha O_{i,k} + \tilde{P}V_{j,k}$ 

```

```

for all  $i, j \in [0, N) \times [0, d)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 

```

FlashAttention-2